A Binary Hilbert Hotel

Black motions, ruby compilation, and the Leavitt nonsoficity proof

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Abstract

For a field $\mathbb{F}$, let

$$R_{\mathbb{F}} = L_{\mathbb{F}}(1,2) = \mathbb{F}\langle D_\ell, D_r, U_\ell, U_r \mid U_a D_b = \delta_{ab},\ D_\ell U_\ell + D_r U_r = 1\rangle, \qquad \Gamma_{\mathbb{F}} = \mathrm{EL}_3(R_{\mathbb{F}}).$$

We present a direct route to the recent theorem that, whenever $\Gamma_{\mathbb{F}}$ has property (T), it is nonsofic; in particular this holds for every finite field. No novelty is claimed. The purpose is to make the exact finite obstruction and its robustification line up as closely as possible.

The warmup is a finite labelled-graph argument. The H-edges are black, the compressor edges are ruby, and the commuting-symmetry edges are sapphire. There is a short exact proof using the finite set of global black symmetries. We then unpack what that proof hides: ruby first cannot lose vertices from black components, and then cannot lose black-equivariant arrows between components. These are the two finite counting statements that survive in the sofic proof.

After two applications of Kun's repair theorem, the finite component set is replaced by black expanders. A stationary Markov-chain lemma upgrades one-sided inequalities to two-sided control; it is used once for component sizes and once for the number of outgoing intertwiner codes. Approximate intertwiners are repaired in a diagonal product graph. Their unique-decoding property is explained by a basic robust-connectivity fact: deleting few edges from an expander leaves one giant connected component, on which agreement propagates. A more-than-half argument then recovers the distinguished sapphire symmetry and yields the final contradiction.

The same binary geometry is drawn first in Thompson's group V, then linearized as the binary Leavitt algebra acting on $\mathbb{F}$-labelled infinite paths. Appendices give a spectral proof of the needed Kun repair theorem and the short three-subgroup property-(T) argument over finite fields of order greater than four.

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1 The theorem and the four-term configuration

The concrete statement is the following.

Theorem 1.1 (Binary Leavitt nonsoficity)**.** *Let $\mathbb{F}$ be a countable field. If*

$$\Gamma_{\mathbb{F}} = \mathrm{EL}_3(L_{\mathbb{F}}(1,2))$$

has property (T), then $\Gamma_{\mathbb{F}}$ is nonsofic. Consequently it is nonsofic for every finite field $\mathbb{F}$.

The construction is encoded by four relations. Inside the ambient group we find a subgroup $H \cong \Gamma_{\mathbb{F}}$ and elements ρ, σ such that

$$\Gamma_{\mathbb{F}} = \langle H, \rho, \sigma \rangle, \qquad \rho H \rho^{-1} \leq H, \qquad [\sigma, H] = 1, \qquad [\rho\sigma\rho^{-1}, H] \neq 1. \tag{1}$$

Thus ruby ρ compresses the black subgroup H one-sidedly; sapphire σ is an H-symmetry; but the transported sapphire motion $\rho\sigma\rho^{-1}$ is not.

The exact finite paradox is simple: an injective self-map of a finite symmetry set is onto. The robust proof has to reconstruct two things that exact finiteness supplied for free:

the orbit picture, and the symmetry-between-orbits picture.

Exact finite model	Sofic/Kazhdan model	Upgrade mechanism
black components Q	black H-expanders Q	Kun repair
$\rho Q \subseteq \pi(Q)$	ρQ mostly lies in $\pi(Q)$	dominant target
finite component cycles force equality	stationary expansion forces size equality	first one-sided-to-two-sided step
exact black-equivariant arrows	decoded approximate intertwiners	product-graph repair
ruby injects outgoing arrows	ruby injects outgoing codes	large-overlap compiler
finite cycles force no loss	stationary expansion forces almost no loss	second one-sided-to-two-sided step
sapphire arrow is in the image	sapphire code is forced into the image	$> 1/2$ lemma
forbidden commutator closes everywhere	forbidden commutator closes almost everywhere	contradict faithfulness

The nonsofic proof is the exact finite proof with "finite counting forbids loss" replaced twice by "stationarity plus expansion forbids persistent loss."

1.1 Provenance

The final rigidity theorem is the specialization of Kun–Thom's 2026 Theorem B needed here [17]. Kun's theorem supplies expander components [9]. Kun–Thom 2019 supplies finite-like almost symmetries in a one-expander setting [13]. OpenAI's 2026 Proposition 2.3 introduced the component-synchronization mechanism used for the first one-sided-to-two-sided step [14]. Alekseev–Thom developed the many-component approximate-intertwiner structure [16]. The presentation below is an expository synthesis of those ingredients.

2 Finite labelled graphs and local correctness

Let $\Lambda = \langle S \rangle$, where $S = S^{-1}$ is finite.

Definition 2.1 (S-graph)**.** *An S-graph on a finite set X consists of permutations*

$$T_s \in \mathrm{Sym}(X) \qquad (s \in S),$$

with $T_{s^{-1}} = T_s^{-1}$. We draw the labelled arc $x \xrightarrow{s} T_s x$. For an S-word w, write wx for the endpoint of its labelled walk from x.

Definition 2.2 (Consistency, faithfulness, correctness)**.** *A vertex x is*

- R-consistent *if every true relation of length at most R closes at x;*
- R-faithful *if no nontrivial word of length at most R closes at x;*
- R-correct *if it is both.*

Equivalently,

$$wx = x \quad \Longleftrightarrow \quad w = 1 \text{ in } \Lambda \qquad (|w| \le R).$$

Definition 2.3 (LEF and sofic)**.** *The group Λ is*

- LEF *if, for every R, there is a finite S-graph in which every vertex is R-correct;*
- sofic *if, for every R and $\delta > 0$, there is a finite S-graph in which at least a $(1-\delta)$-fraction of the vertices are R-correct.*

These are the usual definitions, up to a harmless factor of two in the radius.

We will repeatedly use the bookkeeping principle

consistency builds the finite structure; faithfulness supplies the last contradiction.

3 The exact black–ruby–sapphire warmup

Assume throughout this section that

$$\Gamma = \langle H, \rho, \sigma \rangle, \qquad \rho H \rho^{-1} \le H, \qquad [\sigma, H] = 1, \qquad [\rho\sigma\rho^{-1}, H] \ne 1,$$

and that H is finitely generated.

Fix a finite symmetric black generating set

$$H = \langle b_1, \dots, b_d \rangle.$$

For each i, choose a black word u_i representing $\rho b_i \rho^{-1}$. Enlarge the black generating set once so that, for some $b_\star$,

$$w = [\rho\sigma\rho^{-1}, b_\star] \ne 1.$$

Choose R large enough to see all relations

$$\rho b_i \rho^{-1} = u_i, \qquad \sigma b_i = b_i \sigma,$$

and the nonrelation $w \ne 1$.

Suppose toward contradiction that X is a finite R-correct S-graph, where

$$S = \{b_1^{\pm 1}, \dots, b_d^{\pm 1}, \rho^{\pm 1}, \sigma^{\pm 1}\}.$$

Color the b_i-edges black, the ρ-edges ruby, and the σ-edges sapphire. We write the same symbols for the corresponding permutations of X. By R-consistency, all the displayed bounded relations now hold as exact identities of these permutations.

3.1 The shortest exact proof: the finite black-symmetry set

Define

$$\mathcal{S}_B(X) = \{c \in \mathrm{Sym}(X) : cb_i = b_i c \text{ for all } i\}.$$

This is the finite set of global label-preserving symmetries of the black graph.

The exact commutation identities give

$$\sigma \in \mathcal{S}_B(X).$$

Define the backward ruby compiler

$$F : \mathcal{S}_B(X) \longrightarrow \mathcal{S}_B(X), \qquad F(c) = \rho^{-1} c \rho.$$

Indeed, if c commutes with every black generator, then it commutes with every black word; hence

$$F(c)b_i = \rho^{-1} c \rho b_i = \rho^{-1} c u_i \rho = \rho^{-1} u_i c \rho = b_i F(c).$$

Thus F is a well-defined injection of a finite set into itself, so it is onto. Since $\sigma \in \mathcal{S}_B(X)$, there is $c \in \mathcal{S}_B(X)$ with

$$\rho^{-1} c \rho = \sigma.$$

Necessarily

$$c = \rho \sigma \rho^{-1}.$$

Therefore $\rho\sigma\rho^{-1}$ commutes with $b_\star$ as a permutation of X, so the w-walk closes at every vertex.

But $w \neq 1$ and $|w| \leq R$. The R-faithfulness of every vertex says that the w-walk closes at no vertex. Contradiction.

Theorem 3.1 (Exact one-sided symmetry obstruction)**.** *The four displayed relations imply that* Γ *is not LEF.*

This proof is the right one if brevity is the only objective. To understand the robust proof, however, we should unpack the two finite counting facts hidden inside the single finite set $\mathcal{S}_B(X)$.

3.2 First hidden count: ruby cannot lose component mass

Delete ruby and sapphire and let $Q, R, \ldots$ be the black connected components. The conjugation relations imply that ruby carries a black edge to a prescribed black path. Consequently, for every component Q there is a component $\pi(Q)$ with

$$\rho Q \subseteq \pi(Q). \tag{2}$$

Since ρ is a permutation,

$$|Q| = |\rho Q| \leq |\pi(Q)|.$$

The map π is surjective: given y in a black component R, write $y = \rho x$ and let Q be the black component of x; then $y \in \rho Q \subseteq \pi(Q)$, so $\pi(Q) = R$. Since there are finitely many components, π is a permutation.

Around a π-cycle,

$$|Q| \leq |\pi Q| \leq |\pi^2 Q| \leq \cdots \leq |Q|.$$

Thus all inequalities are equalities, and (2) becomes

$$\boxed{\rho Q = \pi(Q).} \tag{3}$$

This is the exact orbit-level precursor of the first expansion argument.

3.3 Second hidden count: ruby cannot lose black-equivariant arrows

For black components Q, R, define

$$\mathcal{I}(Q,R) = \mathrm{Iso}_B(Q,R)$$

to be the set of bijections $a : Q \to R$ satisfying

$$a(b_i x) = b_i a(x) \qquad \text{for every } i \text{ and } x \in Q.$$

These exact intertwiners compose and invert. Put

$$\mathrm{Out}(Q) = \bigsqcup_R \mathcal{I}(Q,R), \qquad s(Q) = |\,\mathrm{Out}(Q)|.$$

If $a : Q \to R$ is an intertwiner, composition with a gives a bijection

$$\mathrm{Out}(R) \longrightarrow \mathrm{Out}(Q), \qquad c \longmapsto c \circ a.$$

Hence

$$a : Q \to R \quad \Longrightarrow \quad s(Q) = s(R). \tag{4}$$

Sapphire provides a distinguished arrow

$$a_Q = \sigma|_Q \in \mathcal{I}(Q, p(Q)), \qquad p(Q) = \sigma Q.$$

Now use (3). For $a \in \mathcal{I}(\pi Q, \pi R)$ define

$$F_{Q,R}(a) = \rho^{-1} a \rho : Q \to R.$$

The same calculation as above shows that this is an exact black intertwiner, and the map is injective. Since π permutes components, these maps combine to an injection

$$\mathrm{Out}(\pi Q) \hookrightarrow \mathrm{Out}(Q).$$

Therefore

$$s(\pi Q) \le s(Q).$$

Going around a finite π-cycle forces equality everywhere. Thus the ruby compiler fills all of $\mathrm{Out}(Q)$.

In particular the sapphire arrow a_Q lies in the compiler image. Its preimage is an arrow

$$c_Q : \pi Q \to \pi p(Q)$$

with

$$\rho^{-1} c_Q \rho = \sigma|_Q.$$

Therefore

$$c_Q = \rho \sigma \rho^{-1}|_{\pi Q}$$

is black-equivariant. Since every component is some πQ, the transported sapphire motion commutes with the black action everywhere, and the same final faithfulness contradiction follows.

Remark 3.2 (Why keep both exact proofs?)**.** *The symmetry-set proof is much shorter. The component/intertwiner proof is the exact anatomy of the robust argument. It exhibits the two quantities to which expansion will later be applied:*

$$\boxed{|Q| \quad \textit{and} \quad s(Q) = |\,\mathrm{Out}(Q)|.}$$

4 The tree picture: Thompson's group V

The rooted binary tree already contains the whole geometry, except for property (T).

4.1 Rooms and rotations

Let $\mathcal{T}$ be the complete rooted binary tree. A vertex determines the rooted subtree below it; call such a subtree a *room*. A room D has child rooms D^ℓ and D^r.

Thompson's group V consists of finite prefix rearrangements of the boundary of $\mathcal{T}$: partition the infinite paths into finitely many rooms, match two such partitions, and use the canonical rooted-tree identification on each matched pair [2, 10]. Equivalently, V is generated by swaps of two disjoint rooms. Write $V_{\mathsf{D}} \cong V$ for the subgroup supported in D.

The standard left rotation supported in a room D is

$$\mathsf{D}^\ell,\ \mathsf{D}^{r\ell},\ \mathsf{D}^{rr} \longmapsto \mathsf{D}^{\ell\ell},\ \mathsf{D}^{\ell r},\ \mathsf{D}^r.$$

Abstractly this is the data-structures move

$$X(YZ) \longmapsto (XY)Z.$$

Thompson's F is generated, up to left/right and inverse convention, by one such rotation at the root and one one level lower. Thompson's V allows the same finite reparenthesizations together with finite permutations of the hanging rooms.

4.2 The hotel, the rotation, and the outside swap

Choose sibling rooms A, B and a complementary room C, so that

$$\mathsf{A} \sqcup \mathsf{B} \sqcup \mathsf{C}$$

partitions the infinite paths. Put

$$H = V_{\mathsf{A}} \cong V.$$

Think of A as the hotel.

Define $\rho \in V$ by the four-room rearrangement

$$\mathsf{A},\ \mathsf{B},\ \mathsf{C}^\ell,\ \mathsf{C}^r \longmapsto \mathsf{A}^\ell,\ \mathsf{A}^r,\ \mathsf{B},\ \mathsf{C}. \tag{5}$$

This is one ordinary binary-tree rotation viewed at the appropriate scale. Let σ swap the two outside rooms B and C.

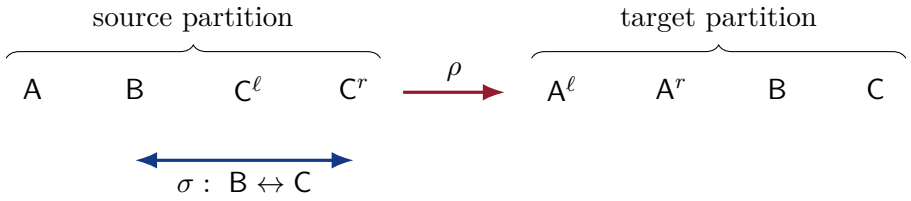


Figure 1: The binary hotel. Ruby packs the hotel into its left child. Sapphire swaps two outside rooms.

The support picture gives

$$\rho H \rho^{-1} = V_{\mathsf{A}^\ell} \leq V_{\mathsf{A}} = H, \qquad [\sigma, H] = 1.$$

Put $\tau = \rho\sigma\rho^{-1}$. Tracing one small room gives

$$\mathsf{A}^{r\ell} \xrightarrow{\rho^{-1}} \mathsf{B}^\ell \xrightarrow{\sigma} \mathsf{C}^\ell \xrightarrow{\rho} \mathsf{B}.$$

Thus τ carries a room inside the hotel outside it. If h is nontrivial and supported in $\mathsf{A}^{r\ell}$, then $\tau h \tau^{-1}$ is supported in B, so $[\tau, h] \neq 1$.

Finally,

$$V = \langle H, \rho, \sigma \rangle.$$

Indeed, $\rho^{-1} H \rho$ gives every rearrangement inside the parent room $\mathsf{A} \sqcup \mathsf{B}$, hence in particular inside B. Conjugation by σ gives rearrangements inside C. Conjugates of σ then swap arbitrary sufficiently small rooms across the cut $(\mathsf{A} \sqcup \mathsf{B}) \mid \mathsf{C}$, and such room swaps generate V.

Thus Thompson's V displays all four relations. The only missing ingredient for nonsoficity is property (T).

5 Linearizing the binary hotel

Fix a field $\mathbb{F}$ and write

$$R = L_{\mathbb{F}}(1, 2), \qquad \Gamma = \mathrm{EL}_3(R).$$

5.1 Path labelings and the binary Leavitt algebra

Let $\partial\mathcal{T} = \{\ell, r\}^{\mathbb{N}}$ be the set of infinite binary paths, and let

$$W_{\mathbb{F}} = \mathbb{F}^{(\partial\mathcal{T})}$$

be the vector space of finitely supported $\mathbb{F}$-labelings of those paths. Define linear operators

$$D_\ell e_x = e_{\ell x}, \qquad D_r e_x = e_{rx},$$

and

$$U_\ell e_{\ell x} = e_x, \quad U_\ell e_{rx} = 0, \qquad U_r e_{rx} = e_x, \quad U_r e_{\ell x} = 0.$$

Thus D_ℓ, D_r put a labeling into the left or right half of the tree, while U_ℓ, U_r extract that half and rescale it back to the whole tree. The relations are

$$U_a D_b = \delta_{ab}, \qquad D_\ell U_\ell + D_r U_r = 1. \tag{6}$$

The second identity says exactly that every labeling is the sum of its left-half and right-half pieces:

$$W_{\mathbb{F}} \cong W_{\mathbb{F}} \oplus W_{\mathbb{F}}.$$

The binary Leavitt algebra is the universal algebra generated by these four operations. It was introduced by Leavitt in 1962 as the prototypical ring for which one free module is isomorphic to two copies of itself [1].

More generally, if α, β are finite binary addresses, the partial room operator

$$E_{\alpha \leftarrow \beta} e_{\beta x} = e_{\alpha x}, \qquad E_{\alpha \leftarrow \beta} e_y = 0 \quad (y \notin \beta\partial\mathcal{T})$$

replaces prefix β by prefix α. It is the word $D_\alpha U_\beta$. Finite prefix rearrangements in V become units which are finite sums of such room operators. In this sense the Leavitt algebra is the linearization of the same prefix calculus underlying V.

5.2 Three registers and the linear rotation

Take three copies of $W_{\mathbb{F}}$, named $\mathsf{A}, \mathsf{B}, \mathsf{C}$. For distinct registers X, Y and $a \in R$, write

$$E_{X \leftarrow Y}(a)$$

for the elementary instruction

$$x_X \leftarrow x_X + a(x_Y),$$

fixing the third register. These instructions generate Γ. Their routing rule is the Steinberg relation

$$[E_{X \leftarrow Y}(a), E_{Y \leftarrow Z}(b)] = E_{X \leftarrow Z}(ab) \tag{7}$$

for distinct X, Y, Z.

The same three-register machine can be installed inside any room D; denote the localized copy by $\Gamma_{\mathsf{D}} \cong \Gamma$. We use only

$$\mathsf{E} \subseteq \mathsf{D} \Longrightarrow \Gamma_{\mathsf{E}} \leq \Gamma_{\mathsf{D}}, \qquad \mathsf{D} \cap \mathsf{E} = \varnothing \Longrightarrow [\Gamma_{\mathsf{D}}, \Gamma_{\mathsf{E}}] = 1.$$

Define the linear rotation by the same support picture (5). In register order $(\mathsf{A}, \mathsf{B}, \mathsf{C})$,

$$\rho = \begin{pmatrix} D_\ell & D_r & 0 \\ 0 & 0 & U_\ell \\ 0 & 0 & U_r \end{pmatrix}. \tag{8}$$

It sends

$$(x_{\mathsf{A}}, x_{\mathsf{B}}, x_{\mathsf{C}}) \longmapsto (D_\ell x_{\mathsf{A}} + D_r x_{\mathsf{B}}, U_\ell x_{\mathsf{C}}, U_r x_{\mathsf{C}}).$$

Lemma 5.1. *The matrix ρ lies in Γ.*

Proof. Let

$$\rho_- = \begin{pmatrix} D_\ell & D_r & 0 \\ 0 & 0 & -U_\ell \\ 0 & 0 & -U_r \end{pmatrix}.$$

The six elementary row operations

$$R_2 \leftarrow R_2 + U_\ell R_1, \quad R_3 \leftarrow R_3 + U_r R_1,$$

$$R_1 \leftarrow R_1 - D_\ell R_2, \quad R_1 \leftarrow R_1 - D_r R_3,$$

$$R_2 \leftarrow R_2 + U_\ell R_1, \quad R_3 \leftarrow R_3 + U_r R_1$$

reduce ρ_-, using (6), to a three-cycle permutation matrix. Hence $\rho_- \in \mathrm{EL}_3(R)$. Finally

$$\rho = \mathrm{diag}(1, -1, -1)\rho_-,$$

and the diagonal factor is elementary because the signed swap

$$\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} = e_{12}(1)e_{21}(-1)e_{12}(1)$$

has square $-I_2$. □

Let σ be the signed elementary swap of the two outside registers B and C.

Proposition 5.2 (The linear binary hotel)**.** *Install*

$$H = \Gamma_{\mathsf{A}} \cong \Gamma$$

inside register A*. Then*

$$\Gamma = \langle H, \rho, \sigma \rangle, \qquad \rho H \rho^{-1} \leq H, \qquad [\sigma, H] = 1, \qquad [\rho\sigma\rho^{-1}, H] \neq 1.$$

Proof. The rotation sends A into A^{ℓ}, so

$$\rho H \rho^{-1} = \Gamma_{\mathsf{A}^{\ell}} \leq \Gamma_{\mathsf{A}} = H.$$

The swap acts only on $\mathsf{B} \cup \mathsf{C}$, so it commutes with H. The same room trace as in V shows that $\rho\sigma\rho^{-1}$ moves $\mathsf{A}^{r\ell}$ to B; conjugating a nonidentity elementary operation supported in $\mathsf{A}^{r\ell}$ therefore moves its support outside A.

For generation, rotate the hotel backward. The group $\rho^{-1} H \rho$ is a local three-register machine on

$$\mathsf{A},\ \mathsf{B}^{\ell},\ \mathsf{B}^{r}.$$

Thus $\langle H, \rho \rangle$ contains arbitrary transfers between A and either half of B. The identity $1 = D_{\ell}U_{\ell} + D_{r}U_{r}$ assembles the two half-register transfers into arbitrary transfers between A and B. Conjugating by σ gives all transfers between A and C, and routing through A using (7) gives all transfers between B and C. □

6 Property (T) over finite fields

The main theorem only needs property (T) as an input. For finitely generated rings it follows from Ershov–Jaikin-Zapirain [6]. Over finite fields of order $q > 4$ there is a much shorter three-subgroup proof.

Let R be any finitely generated unital associative $\mathbb{F}_q$-algebra. Choose a finite-dimensional subspace $A \subseteq R$ containing 1 and a finite algebra-generating set, and put

$$C = \operatorname{span}_{\mathbb{F}_q}(AA).$$

For $x_{ij}(a) = I + aE_{ij}$, set

$$H_1 = x_{12}(A), \qquad H_2 = x_{23}(A), \qquad H_3 = x_{31}(A).$$

Each H_i is a finite elementary abelian q-group.

Lemma 6.1 (Three-subgroup criterion)**.** *Let a group J be generated by finite subgroups H_1, H_2, H_3. Suppose every pairwise join $\langle H_i, H_j \rangle$ is finite and the averaging operator*

$$A_{ij} = \frac{1}{2}(P_i + P_j)$$

has spectral gap $\gamma > 1/4$ away from the common invariant vectors, where P_i is averaging over H_i. Then J has property (T).

Proof. Put

$$\Delta_i = I - P_i, \qquad \Delta = \frac{1}{3}(\Delta_1 + \Delta_2 + \Delta_3), \qquad \Delta_{ij} = \frac{1}{2}(\Delta_i + \Delta_j).$$

Since $\Delta_i^2 = \Delta_i$,

$$\Delta^2 - \kappa\Delta = \frac{4}{3} \operatorname*{avg}_{i<j} \left(\Delta_{ij}^2 - \left(\frac{1}{4} + \frac{3}{4}\kappa \right) \Delta_{ij} \right).$$

Choose $\kappa > 0$ with $\frac{1}{4} + \frac{3}{4}\kappa < \gamma$. Each term is positive semidefinite in every unitary representation of the finite pairwise join. Thus Δ has a uniform spectral gap above its kernel, which is the J-fixed space. □

For instance, $\langle H_1, H_2 \rangle$ has coordinates

$$(a, c, b) \in A \times C \times A$$

with multiplication

$$(a, c, b)(a', c', b') = (a + a', c + c' + ab', b + b').$$

This is a finite generalized Heisenberg group. The standard orthogonality-matrix calculation gives

$$\gamma \geq \frac{1}{2}(1 - q^{-1/2}).$$

Indeed, every nonzero central mode sees a nonzero bilinear form $(a, b) \mapsto \lambda(ab)$; because C is spanned by products, the corresponding normalized matrix contains at least one nondegenerate $q \times q$ orthogonality block and therefore has operator norm at most $q^{-1/2}$.

For $q > 4$, the displayed gap exceeds $1/4$. The three subgroups generate $\mathrm{EL}_3(R)$: Steinberg commutators produce all six root directions with coefficients in A, and repeated sums and commutators enlarge the coefficient set to the subring generated by A, namely R.

Proposition 6.2. *If R is a finitely generated unital associative $\mathbb{F}_q$-algebra and $q > 4$, then $\mathrm{EL}_3(R)$ has property (T).*

In particular this applies to $R = L_{\mathbb{F}_q}(1, 2)$. For the remaining finite fields, property (T) follows from the general theorem of Ershov–Jaikin-Zapirain.

7 One-sided symmetry rigidity

We now state the robust theorem whose special case proves nonsoficity.

Theorem 7.1 (One-sided symmetry rigidity)**.** *Let $H \leq \Gamma$ be finitely generated groups with property (T). Suppose*

$$\Gamma = \langle H, \rho, \sigma \rangle, \qquad \rho H \rho^{-1} \leq H, \qquad [\sigma, H] = 1.$$

If Γ is sofic, then

$$[\rho\sigma\rho^{-1}, H] = 1.$$

The proof below is a finite graph presentation of the needed specialization of Kun–Thom Theorem B. Its structure is now exactly parallel to the anatomical finite proof:

Exact finite proof	Robust proof
black components	repaired black expanders
$\rho Q \subseteq \pi Q$	ρQ has a dominant target πQ
component sizes cannot increase around a cycle	component-size potential cannot drift on an ambient expander
exact intertwiners $\mathrm{Iso}_B(Q, R)$	decoded approximate intertwiners $\mathcal{I}(Q, R)$
ruby injects $\mathrm{Out}(\pi Q)$ **into** $\mathrm{Out}(Q)$	ruby compiler injects outgoing codes through large overlaps
$s(\pi Q) \leq s(Q)$ **cannot persist around a cycle**	outgoing-code potential cannot drift on an ambient expander
sapphire arrow belongs to the image	$> 1/2$ forces sapphire code into the image

8 Kun repair and the two expander decompositions

We use the following finite local form of Kun's theorem.

Theorem 8.1 (Kun repair theorem, finite form)**.** *Let $\Lambda = \langle S \rangle$ have property (T), with $S = S^{-1}$ finite. There is a constant $h_\Lambda > 0$ such that for every repair budget $\eta > 0$ there are a radius L and a tolerance $\delta > 0$ with the following property. If a finite S-graph has at least $(1 - \delta)|Y|$ vertices which are L-consistent with Λ, then one may change at most $\eta|Y|$ generator values so that every connected component of the repaired graph is either a singleton or has Cheeger constant at least h_Λ.*

The repaired components are exactly invariant under the repaired generator permutations. They need not be genuine finite Schreier graphs: the repair creates expanders, not a global action of Λ. Appendix A gives a spectral proof of this finite form.

Assume toward contradiction that theorem 7.1 fails. Fix a symmetric generating set $S = S^{-1}$ of H, enlarged so that for some $h_0 \in S$,

$$w = [\rho\sigma\rho^{-1}, h_0] \neq 1.$$

Choose one very accurate finite sofic model of Γ, with word radius and error tolerance small enough for every bounded relation, Kun repair, product-graph repair, and final word test below.

Apply theorem 8.1 first to the ambient Γ-graph. After a small repair it is a disjoint union of uniform ambient expanders. Put into an exceptional set every initially incorrect vertex, every vertex touched by the repair, and the bounded neighborhoods of these vertices which later word calculations may reach. By averaging, all but a small fraction of the total vertex mass lies in ambient components on which this exceptional set has small density. Fix one such clean ambient component A. It is a nontrivial uniform expander.

From this point onward, $o(1)$ is shorthand for a quantity that can be made smaller than any prescribed constant by choosing this one finite model sufficiently accurate after all radii and repair scales have been fixed. No ultraproduct or limiting object is used; a fully quantitative version replaces the occurrences of $o(1)$ by a hierarchy of explicit powers of one small error parameter.

Inside A, ignore ruby and sapphire and apply Kun again to the black H-graph. After another small repair,

$$A = Q_1 \sqcup \cdots \sqcup Q_m, \tag{9}$$

where each nonsingleton Q_i is a uniform black expander. We retain two graph structures on the same vertex set:

- the first-stage ambient graph, used for global expansion;
- the second-stage black graph, whose connected components are the Q_i.

They differ from the original sofic model only on small exceptional sets.

9 The stationary one-sided-to-two-sided lemma

The analytic core is a basic finite Markov-chain fact. It is independent of groups and property (T).

Lemma 9.1 (Stationary one-sided-to-two-sided lemma)**.** *Let P be a finite Markov chain with stationary distribution π. Let P^* be its time reversal and*

$$P_{\mathrm{sym}} = \frac{1}{2}(P + P^*)$$

its additive reversibilization. Suppose P_{sym} has Poincaré gap at least $\lambda > 0$.

Sample

$$X \sim \pi, \qquad Y \sim P(X, \cdot).$$

If $F : V \to [0,1]$ satisfies

$$\mathbb{E}[(F(Y) - F(X))_+] \le \eta,$$

then

$$\mathrm{Var}_\pi(F) \le \frac{\eta}{\lambda}.$$

Proof. Stationarity gives

$$\mathbb{E}[F(Y) - F(X)] = 0.$$

Thus expected upward variation equals expected downward variation, and

$$\mathbb{E}|F(Y) - F(X)| = 2\mathbb{E}[(F(Y) - F(X))_+] \le 2\eta.$$

Since $0 \le F \le 1$,

$$(F(Y) - F(X))^2 \le |F(Y) - F(X)|.$$

Therefore

$$\mathcal{E}_{P_{\mathrm{sym}}}(F, F) = \frac{1}{2}\mathbb{E}(F(Y) - F(X))^2 \le \eta.$$

The Poincaré inequality gives

$$\lambda \,\mathrm{Var}_\pi(F) \le \mathcal{E}_{P_{\mathrm{sym}}}(F, F) \le \eta.$$

□

The proof is the entire story:

stationarity turns one-sided small drift into two-sided small variation, and spectral gap turns small variation into near constancy.

Corollary 9.2 (Multiplicative form)**.** *In the setting of theorem 9.1, let $f : V \to (0, \infty)$ and let m be a median of f under π. Suppose*

$$\mathbb{P}[f(Y) > (1+\gamma)f(X)] \le \delta.$$

Put

$$F(x) = \frac{f(x)}{f(x)+m}.$$

Then

$$\mathrm{Var}_\pi(F) \le \frac{\gamma/4+\delta}{\lambda}.$$

Consequently, for every $0 < \tau < 1/2$, outside a set of π-measure at most

$$\frac{3(\gamma/4+\delta)}{\lambda\tau^2},$$

one has

$$\frac{1-2\tau}{1+2\tau} \le \frac{f(x)}{m} \le \frac{1+2\tau}{1-2\tau}.$$

Proof. On a good transition, monotonicity gives

$$(F(Y)-F(X))_+ \le \frac{(1+\gamma)a}{m+(1+\gamma)a} - \frac{a}{m+a} \le \frac{\gamma a m}{(a+m)^2} \le \frac{\gamma}{4},$$

where $a = f(X)$. On a bad transition the increase is at most 1. Apply theorem 9.1. Since $1/2$ is a median of F, its mean differs from $1/2$ by at most a constant multiple of $\sqrt{\mathrm{Var}(F)}$; hence

$$\mathbb{E}\left(F - \frac{1}{2}\right)^2 \le 3\,\mathrm{Var}(F).$$

Markov's inequality and the identity $f/m = F/(1-F)$ finish the proof. □

In our application, P is an average of finitely many generator permutations: the black directions, sapphire, and the forward ruby direction. Such an average is doubly stochastic, so the uniform distribution on A is stationary. The symmetrization restores the inverse directions and is comparable to the ambient repaired generator graph, hence has a uniform gap.

10 First use: component synchronization

10.1 A dominant-target lemma

Lemma 10.1 (Dominant target)**.** *Let Q be an h-expander, and partition its vertices into cells. If b edges of Q cross between distinct cells, then some cell C satisfies*

$$|Q \setminus C| \le \frac{2b}{h}.$$

Proof. If the largest cell has more than half the vertices, apply expansion to its complement. Otherwise apply expansion to every cell and sum; every crossing edge is counted at most twice. □

10.2 Sapphire is already two-sided

Transport the black expander structure on Q through σ. The local relations $\sigma s = s\sigma$, together with the small discrepancy between original and repaired black edges, imply that only a small number of transported black edges cross the partition (9). By theorem 10.1, almost all of σQ lies in one black component, call it $p(Q)$.

Run the same argument for σ^{-1}. Outside a small total mass of components, the forward and backward dominant targets agree. Thus

$$|\sigma Q \triangle p(Q)| \ll |Q|, \qquad |p(Q)| = (1 + o(1))|Q|,$$

and $\sigma|_Q$ is an approximate black-equivariant bijection. No stationary balancing is needed because sapphire is reversible from the start.

10.3 Ruby is one-sided

For each $s \in S$, fix an S-word w_s representing $\rho s \rho^{-1} \in H$. Transport the black expander on Q through ρ. Away from a small exceptional set, a transported s-edge agrees with the bounded black path w_s. Thus few transported black edges cross the partition (9), and theorem 10.1 gives a target $\pi(Q)$ with

$$|\rho Q \setminus \pi(Q)| \ll |Q|. \tag{10}$$

Initially this yields only

$$|\pi(Q)| \gtrsim |Q|. \tag{11}$$

There is no corresponding statement from ρ^{-1}.

Define on vertices of A

$$f(x) = \frac{1}{|Q(x)|},$$

where $Q(x)$ is the black component containing x. Along black edges, f is constant except at repair errors. Along sapphire edges, the two-sided matching approximately preserves it. Along a good forward ruby edge from Q into $\pi(Q)$, (11) gives

$$f(\rho x) \le (1 + o(1)) f(x).$$

Apply theorem 9.2 to the stationary generator walk on the ambient expander. We conclude that, outside a small set,

$$f(\rho x) = (1 + o(1)) f(x).$$

Hence, after discarding components carrying small total mass,

$$|\pi(Q)| = (1 + o(1))|Q|.$$

Together with (10), this gives

$$|\rho Q \triangle \pi(Q)| = o(|Q|). \tag{12}$$

Thus ruby approximately permutes the black components. This is the robust form of the exact count

$$\rho Q \subseteq \pi Q \quad \Longrightarrow \quad \rho Q = \pi Q.$$

11 Approximate intertwiners and robust decoding

Component matching reconstructs only the orbit picture. The exact proof also used the finite family of black-equivariant arrows between components. We now build their robust replacement.

For black expanders Q, R, an approximate intertwiner is a partial bijection

$$a : Q \dashrightarrow R$$

with small source loss, small range loss, and small black-equivariance defect. More explicitly, put

$$\lambda_Q(a) = \frac{|Q \setminus \operatorname{dom} a|}{|Q|}, \qquad \lambda_R(a) = \frac{|R \setminus \operatorname{ran} a|}{|R|},$$

and

$$\Delta_S(a) = \frac{1}{|Q|} \left|\{(x, s) \in Q \times S : x \notin \operatorname{dom} a, sx \notin \operatorname{dom} a, \text{ or } a(sx) \neq sa(x)\}\right|.$$

Undefined values count as failures.

11.1 The graph-of-a-map repair

Consider the diagonal black graph on $Q \times R$:

$$(x, y) \overset{s}{\mapsto} (sx, sy).$$

The graph of a,

$$B_a = \{(x, a(x)) : x \in \operatorname{dom} a\},$$

has boundary controlled by $\Delta_S(a)$. Thus approximate equivariance is the statement that B_a is a low-boundary matching in the product graph.

The product graph need not itself be an expander. Instead, property (T) supplies the same finite-radius spectral sharpening used in Kun's theorem: a low-boundary set can be changed slightly to one with arbitrarily smaller relative boundary. After sharpening, row and column cleanup turns the set back into the graph of a partial bijection.

Proposition 11.1 (Intertwiner repair)**.** *For every sufficiently small target error $\eta > 0$, there are a local consistency radius and an initial error threshold, depending only on H and its generating set, with the following property. If Q, R are black expanders which are locally H-consistent at the required radius and $a : Q \dashrightarrow R$ has sufficiently small source loss, range loss, and equivariance defect, then one may change a on*

$$O(\Delta_S(a)|Q|) + \eta|Q|$$

source points to obtain an intertwiner a' with

$$\lambda_Q(a'), \lambda_R(a'), \Delta_S(a') \leq \eta.$$

Proof sketch. Apply sparse-set sharpening to $B_a \subseteq Q \times R$. The resulting set U is close to B_a and has extremely small diagonal boundary. Let $m_1(x)$ and $m_2(y)$ be its row and column multiplicities. Discrete coarea plus expansion of Q and R shows that only a small number of rows and columns are empty or overfull. Delete surplus points and match the remaining empty rows with empty columns. The resulting set is the graph of a partial bijection and inherits small boundary, hence small equivariance defect. □

11.2 The natural expander clustering fact

The unique-decoding step rests on a very generic expander statement.

Lemma 11.2 (Robust connectivity of an expander)**.** *Let $G = (V, E)$ be a d-regular graph of conductance at least ϕ. Delete a set $F \subseteq E$ of edges. If*

$$|F| < \frac{1}{2}\phi d|V|,$$

then the remaining graph has a connected component C satisfying

$$|V \setminus C| \leq \frac{|F|}{\phi d}.$$

Proof. If every remaining component had size at most $|V|/2$, expansion of every component and double counting of deleted edges would give

$$2|F| \geq \phi d|V|,$$

a contradiction. Thus one remaining component C has more than half the vertices. Put $U = V \setminus C$. Every edge of ∂U was deleted, so

$$\phi d|U| \leq |\partial U| \leq |F|.$$

□

Now let $a, c : Q \dashrightarrow R$ be two good approximate intertwiners. Delete every black edge of Q on which either map fails equivariance, together with edges touching undefined points. Only a small fraction of edges is deleted. On every surviving edge $x \xrightarrow{s} y$,

$$a(y) = sa(x), \qquad c(y) = sc(x).$$

Since s acts bijectively on R,

$$a(x) = c(x) \quad \Longleftrightarrow \quad a(y) = c(y).$$

Thus the agreement bit is constant on every surviving connected component. By theorem 11.2, there is one giant component. Hence a and c either agree on almost all of Q or disagree on almost all of Q.

Corollary 11.3 (Expander agreement dichotomy)**.** *Two sufficiently good approximate intertwiners with the same source and target are either $O(\eta/\phi)$-close in Hamming distance or $1 - O(\eta/\phi)$-far apart.*

This is the coding phenomenon. Approximate intertwiners separate into tight Hamming clusters with a large forbidden middle range.

11.3 Intertwiner codes

Call two sufficiently accurate intertwiners with the same endpoints equivalent if they are close. By theorem 11.3, the equivalence classes form a finite set

$$\mathcal{I}(Q, R)$$

of *intertwiner codes*. Raw composition and inverse slightly enlarge the defects; repair again using theorem 11.1, then take the code. The decoding gap makes the result independent of all repair choices.

Put

$$\mathrm{Out}(Q) = \bigsqcup_R \mathcal{I}(Q,R), \qquad s(Q) = |\,\mathrm{Out}(Q)|.$$

If $a \in \mathcal{I}(Q,R)$, composition with a gives a bijection

$$\mathrm{Out}(R) \longrightarrow \mathrm{Out}(Q), \qquad c \longmapsto c \circ a,$$

with inverse composition by a^{-1}. Thus

$$a : Q \to R \quad \Longrightarrow \quad s(Q) = s(R). \tag{13}$$

Sapphire supplies a distinguished code

$$a_Q \in \mathcal{I}(Q, p(Q)),$$

so

$$s(p(Q)) = s(Q). \tag{14}$$

12 The ruby compiler and the second one-sided-to-two-sided step

For a retained component Q, put $P_Q = \rho Q$ and give P_Q the transported black labels. By (12), the identity on the large overlap

$$P_Q \cap \pi(Q)$$

is an approximate intertwiner between P_Q and $\pi(Q)$.

Take a code represented by

$$a : \pi(Q) \dashrightarrow \pi(R).$$

Transport it through the two large overlaps:

$$P_Q \dashrightarrow \pi(Q) \xrightarrow{a} \pi(R) \dashrightarrow P_R.$$

Because transported black label s means the bounded black word $\rho s \rho^{-1} = w_s$, the resulting partial map is approximately equivariant. Repair and take its code. This defines an injective ruby compiler

$$F_{Q,R} : \mathcal{I}(\pi(Q), \pi(R)) \hookrightarrow \mathcal{I}^{\rho}(Q,R).$$

It respects composition and inverse. Let J be the compiler image; then J is closed under composition and inverse.

The component matching π misses only small total vertex mass. Components joined by an intertwiner code have comparable sizes, so, outside a small collection of intertwiner classes, almost every target of an arrow lies in the range of π. For such a component Q, every code leaving $\pi(Q)$ toward a target in the range of π is fed injectively through the compiler. Since all reachable targets support the same number of outgoing codes, this gives a one-sided estimate

$$s(\pi(Q)) \le (1 + o(1)) s(Q). \tag{15}$$

Define on vertices of A

$$g(x) = s(Q(x)).$$

Along black edges, g is constant. Along sapphire edges it is preserved by (14). Along good forward ruby edges, (15) gives a one-sided multiplicative bound. Apply theorem 9.2 a second time. We obtain

$$s(\pi(Q)) = (1 + o(1))s(Q)$$

for all components outside small total mass. Therefore the injective compiler image occupies almost all outgoing codes; in particular it occupies strictly more than half.

This is the robust form of the exact count

$$s(\pi Q) \le s(Q) \quad \Longrightarrow \quad s(\pi Q) = s(Q)$$

around a finite component cycle.

13 More than half recovers sapphire

Lemma 13.1 (More than half)**.** *Let J be a collection of intertwiner codes closed under composition and inverse. Let $a : Q \to R$ be a code. If*

$$|\operatorname{Out}_J(Q)| > \frac{1}{2}|\operatorname{Out}(Q)|, \qquad |\operatorname{Out}_J(R)| > \frac{1}{2}|\operatorname{Out}(R)|,$$

then $a \in J$.

Proof. Composition with a is a bijection $\operatorname{Out}(R) \to \operatorname{Out}(Q)$. The image of the $> 1/2$ subset at R and the $> 1/2$ subset at Q must intersect. Thus some $b \in J$ has $b \circ a \in J$, and

$$a = b^{-1}(b \circ a) \in J.$$

□

Apply the lemma to the sapphire code after transporting it through ruby. At both endpoints the compiler image is more than half full, so the transported sapphire code lies in the compiler image. Unwinding the definitions gives approximate intertwiners

$$b_Q : \pi(Q) \dashrightarrow \pi(p(Q))$$

whose ruby transport agrees, up to a small error, with $\rho\sigma\rho^{-1}$ on Q.

Because the source and target components are distinct and the component matchings are injective on the retained family, these maps patch into one partial bijection on almost all of A. It is approximately black-equivariant and differs from $\rho\sigma\rho^{-1}$ on only a small fraction of vertices. Consequently, for every black generator s,

$$\rho\sigma\rho^{-1}s(x) = s\rho\sigma\rho^{-1}(x)$$

for almost every $x \in A$.

Take $s = h_0$. The original word

$$w = [\rho\sigma\rho^{-1}, h_0]$$

therefore fixes almost every vertex of A. But A was chosen clean, and local faithfulness of the sofic model says that this fixed nontrivial word fixes only a tiny fraction of its vertices. Contradiction. This proves theorem 7.1.

14 Application to the binary Leavitt group

Let $R = L_{\mathbb{F}}(1, 2)$ and $\Gamma = \mathrm{EL}_3(R)$. By theorem 5.2, the localized subgroup $H \cong \Gamma$ and the elements ρ, σ satisfy the four relations (1). If Γ has property (T), then so does H. If Γ were sofic, theorem 7.1 would force

$$[\rho\sigma\rho^{-1}, H] = 1,$$

contradicting theorem 5.2. This proves theorem 1.1. For finite fields, property (T) follows from Ershov–Jaikin-Zapirain; for $q > 4$ it also follows from theorem 6.2.

15 The proof in one page

The exact finite proof has a short form and an anatomical form.

- *Short form.* The finite set of black symmetries is invariant under backward ruby conjugation. The resulting injection is onto, so sapphire has a ruby preimage; that preimage is $\rho\sigma\rho^{-1}$.
- *Anatomical form.* Ruby first cannot lose component vertices, and then cannot lose exact intertwiners leaving a component. Both one-sided inequalities become equalities around finite cycles.

The robust proof follows the anatomical form.

1. Kun twice: one ambient expander containing many black expanders.
2. Dominant targets: sapphire is two-sided; ruby is initially one-sided.
3. First stationary balancing: component sizes become two-sidedly matched.
4. Product-graph repair: approximate intertwiners become finite codes.
5. Robust connectivity: after deleting bad edges, agreement propagates on a giant component, giving unique decoding.
6. Ruby compiler: outgoing code counts satisfy a one-sided inequality.
7. Second stationary balancing: the compiler image becomes more than half full.
8. More-than-half: the sapphire code survives ruby transport.
9. The forbidden commutator fixes almost everything, contradicting local faithfulness.

The single conceptual sentence is

one-sided compression cannot destroy a symmetry in a finite or sofic Kazhdan model.

A A spectral proof of the Kun repair theorem

This appendix gives an expository proof of theorem 8.1. The route is

$$\text{Ozawa SOS} \Longrightarrow \text{lazy-walk energy decay} \Longrightarrow \text{sparse-set sharpening}$$
$$\Longrightarrow \text{minimum-set peeling} \Longrightarrow \text{trimming and permutation completion.}$$

The theorem is Kun's [9]; the final trimming shortcut is due to Saranurak–Wang [12] and postdates Kun's original work.

Fix $\Lambda = \langle S \rangle$, $S = S^{-1}$, with property (T). On a finite d-regular S-graph write

$$M = \frac{1}{d}\sum_{s\in S} P_s, \qquad \Delta = I - M, \qquad \mathcal{E}(f) = \langle f, \Delta f\rangle.$$

Then

$$\mathcal{E}(f) = \frac{1}{d}\sum_{\{x,y\}\in E} (f(x)-f(y))^2, \qquad \mathcal{E}(\mathbf{1}_T) = \frac{|\partial T|}{d}.$$

A.1 Ozawa's local spectral certificate

Ozawa's finite-support formulation of property (T) gives $\kappa > 0$ and finitely supported $\xi_1, \dots, \xi_m \in \mathbb{R}[\Lambda]$ such that

$$\Delta^2 - \kappa\Delta = \sum_j \xi_j^* \xi_j. \tag{16}$$

Choose R_0 large enough to cover every word identity appearing when (16) is expanded. If f is supported on vertices whose radius-R_0 neighborhoods are R_0-consistent with Λ, then the same quadratic-form identity holds in the finite graph, so

$$\|\Delta f\|_2^2 \geq \kappa \mathcal{E}(f). \tag{17}$$

Let

$$W = I - \frac{1}{2}\Delta = \frac{1}{2}(I+M), \qquad q = 1 - \frac{\kappa}{2} < 1.$$

Since $0 \preceq \Delta \preceq 2I$,

$$\begin{aligned}\mathcal{E}(Wf) &= \mathcal{E}(f) - \langle f, \Delta^2 f\rangle + \frac{1}{4}\langle f, \Delta^3 f\rangle \\ &\leq \mathcal{E}(f) - \frac{1}{2}\langle f, \Delta^2 f\rangle \\ &\leq q\mathcal{E}(f).\end{aligned}$$

Thus every lazy-walk step removes a constant fraction of Dirichlet energy wherever the bounded-radius certificate is valid.

A.2 Sparse-set sharpening

Proposition A.1 (Sparse-set sharpening)**.** *There are constants $h, C > 0$ depending only on (Λ, S) such that for every $\alpha > 0$ there is a radius $R(\alpha)$ with the following property. If every vertex of T is $R(\alpha)$-consistent and*

$$|\partial T| \leq h|T|,$$

then there is U with

$$|U \triangle T| \leq C|\partial T|, \qquad |\partial U| \leq \alpha|U|.$$

Proof. Put $f_j = W^j \mathbf{1}_T$. Choose an integer t, and take $R(\alpha) \geq R_0 + 2t$. If every vertex of T is $R(\alpha)$-consistent, then every point reached in at most t steps retains enough radius for (17). Hence

$$\mathcal{E}(f_j) \leq q^j \mathcal{E}(\mathbf{1}_T) \qquad (j \leq t).$$

Also

$$f_{j+1} - f_j = -\frac{1}{2}\Delta f_j, \qquad \Delta^2 \preceq 2\Delta,$$

so

$$\|f_{j+1} - f_j\|_2 \leq 2^{-1/2} q^{j/2} \sqrt{\mathcal{E}(\mathbf{1}_T)}.$$

Summing the geometric series gives

$$\|f_t - \mathbf{1}_T\|_2^2 \leq C_0 \mathcal{E}(\mathbf{1}_T) = \frac{C_0}{d}|\partial T|.$$

For $\theta \in [1/3, 2/3]$, put $U_\theta = \{f_t > \theta\}$. Every mismatched vertex contributes at least $1/9$ to the squared distance, so

$$|U_\theta \triangle T| \leq C_1 |\partial T|.$$

The usual coarea/Cauchy–Schwarz sweep gives some θ with

$$|\partial U_\theta| \leq C_2 \sqrt{|T| \mathcal{E}(f_t)} \leq C_3 q^{t/2} \sqrt{|T||\partial T|}.$$

Choose h so small that the symmetric-difference estimate gives $|U_\theta| \geq |T|/2$, then choose $t = t(\alpha)$ so large that the last display is at most $\alpha |U_\theta|$. □

This is stronger than ordinary Cheeger rounding. Property (T) first manufactures a much lower-energy function without letting it move far from the original indicator; thresholding is only the final rounding step.

A.3 Minimum sparse-set peeling

Fix a repair budget η. Choose α and then the local error tolerance δ sufficiently small. Remove the vertices which are not $R(\alpha)$-consistent. In the remaining graph, while there is a nonempty set T with

$$|\partial T| < h|T|,$$

choose one of minimum cardinality, sharpen it to U, and peel off the part of U still present.

Minimality implies that every peeled piece is a nearly constant expander in the original graph: if A is at most half the piece, then $|A| < |T|$, so the minimality of T forces $|\partial A| \geq h|A|$. The terminal remainder has the same property by the stopping rule.

Only a small number of edges cross between pieces. When a piece is peeled, every edge from it to a later piece crosses the sharpened cut ∂U, whose size is at most $\alpha|U|$. Each vertex is charged only once, so there is no logarithmic loss.

A.4 Trimming and permutation completion

Saranurak–Wang's trimming theorem turns a nearly ϕ-expander with small external boundary into a genuine induced expander after deleting $O(|E(P, V \setminus P)|/\phi)$ vertices. Apply it to every peeled piece whose outer boundary is small. Pieces with too much outer boundary, and all vertices removed by trimming, have small total mass. The remaining cores are uniform expanders and have only a small total number of crossing generator arcs.

Finally repair each generator permutation so that every core and every discarded singleton is invariant. For a permutation and a partition, the number of arrows leaving a part equals the number of missing incoming arrows. Keep all internal arrows and match the two deficit sets arbitrarily. This changes exactly the crossing arcs, within the prescribed repair budget, and cannot decrease internal expansion. This proves theorem 8.1.

B The finite-field property-(T) calculation in more detail

We record the elementary Heisenberg-link estimate used in theorem 6.2. Let A and $C = \mathrm{span}(AA)$ be finite-dimensional $\mathbb{F}_q$-spaces as above. The pairwise join $\langle H_1, H_2\rangle$ is the group

$$N = A \times C \times A$$

with multiplication

$$(a, c, b)(a', c', b') = (a + a', c + c' + ab', b + b').$$

The two root subgroups are

$$A \times 0 \times 0, \qquad 0 \times 0 \times A.$$

Decompose the regular representation according to additive characters of the central group C. At the trivial central mode, the two-subgroup averaging operator has no smaller gap than claimed. At a nontrivial mode $\lambda \in C^*$, the relevant normalized matrix is

$$(T_\lambda f)(b) = \frac{1}{|A|} \sum_{a \in A} \chi(\lambda(ab)) f(a),$$

where χ is a fixed nontrivial additive character of $\mathbb{F}_q$.

The bilinear form

$$\beta_\lambda(a, b) = \lambda(ab)$$

is nonzero, because C is spanned by products. After choosing bases, A splits into radical directions and a nondegenerate part of rank at least one. On the nondegenerate part, T_λ is a tensor product containing a normalized $q \times q$ orthogonality matrix. Hence

$$\|T_\lambda\| \le q^{-1/2}.$$

The nontrivial eigenvalues of the two-subgroup averaging operator are therefore bounded above by

$$\frac{1}{2}(1 + q^{-1/2}),$$

so its gap is at least

$$\frac{1}{2}(1 - q^{-1/2}).$$

For $q > 4$ this exceeds $1/4$, and theorem 6.1 gives property (T).

The argument uses only that A is finite-dimensional, contains 1, and generates the algebra. Thus it proves theorem 6.2 for every finitely generated unital associative $\mathbb{F}_q$-algebra, not merely for the Leavitt algebra or for commutative algebras.

C Historical notes and references

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